

Q-ORBIT

Fail-Closed Finite-Key Screening for a Satellite QKD Concept

A controlled engineering study for generating, verifying, managing, and delivering quantum-derived keys - while mission data remains on authenticated encrypted classical channels.

PHASE 1: COMPLETE

CONDITIONAL PASS

ZERO KEY RELEASED

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Conceptual illustration - not to scale

Research Need

Satellite QKD is promising, but a positive software output is not proof of physical security. Q-Orbit asks what the frozen finite-key model supports, which assumptions remain hidden, and where evidence is still missing.

CORE PRINCIPLE

Fail closed: an unmapped device effect remains open; no invented penalty is substituted.

Research Scope

Protocol profile: efficient BB84, weak coherent pulses, one signal intensity and two decoy intensities (one vacuum).

Model chain: orbit and geometry -> optical loss -> detection -> QBER -> finite-key margin.

Evidence class: modeled and deterministic; not measured.

Current Project Status

Manuscript V1.0-RC2: COMPLETE

Prototype: BUILT - software/research only

Website: IN PROGRESS - exhibition interface

Physical characterization: NOT EXECUTED

Hardware-in-loop: BLOCKED

Key release: QUARANTINED / ZERO RELEASED

System Concept

LEO SATELLITE

Optical source / terminal



FREE-SPACE LINK

Modeled channel



GROUND TERMINAL

Detection and sifting



SECURE KEY LAYER

Authentication + KMS

Method

- 01 Freeze the V0.16 reference fixture and canonical values.
- 02 Reproduce the baseline and enforce regression invariants.
- 03 Re-optimize the integration half-window at every evaluated point.
- 04 Run a 41 x 41 deterministic two-parameter screen.
- 05 Rank local numerical response around the baseline.
- 06 Map 15 device imperfections to proof and characterization needs.
- 07 Apply claim-control labels and preserve open blockers.

Verified Theoretical Outputs

SIGNED BASELINE MARGIN

41,338.62418456675 bits

Modeled fixture margin - not a released key

CANDIDATE OUTPUT

41,338 bits

Floored theoretical candidate; quarantined

OPTIMAL HALF-WINDOW

102 s

GRID MEDIAN

-2,624.946810258186 bits

GRID MINIMUM

-3,828.414517626367 bits

GRID MAXIMUM

+142,540.7481180454 bits

GRID PARTITION

568 / 1,113

Positive / nonpositive; not a probability

12/12 regression tests PASS | 20/20 audit checks PASS

Architecture and Boundaries

Layer 0 - Frozen fixture

Immutable numerical regression reference.

Layer 1 - Hardened finite-key analysis

Proof profile, assumption ledger, epsilon accounting, and validators.

Layer 2 - Device-imperfection extensions

Proof-compatible parameters remain symbolic until characterized.

Layer 3 - Characterization composition

Confidence intervals and robust operating envelopes; currently blocked.

BOUNDARY

The frozen numerical fixture assumes a downlink for analysis; the flight/terminal architecture has not yet selected or validated the physical optical-link direction.

EVIDENCE BOUNDARY

INTERIM THEORETICAL SUBMISSION - NOT PHYSICALLY VALIDATED. No mission-success, availability, hardware-readiness, deployment, implementation-security, or released-key claim is made. REQ-01 through REQ-05 remain open and gate independent end-to-end re-execution and provenance claims.